DEPI – Digital Egypt Pioneers Initiative

Data Science & Artificial Intelligence Track

PROJECT PLANNING & MANAGEMENT

Planning Report

Project Proposal, Roles, Risks & KPIs

SMEs Risk Prediction — Machine Learning Model

Machine Learning & AI Team | DEPI 2026

1. Project Proposal

1.1 Overview

The **SME Intelligent Credit** Scoring System is a sophisticated Data Science solution designed to evaluate the financial health of Small and Medium Enterprises (SMEs). Utilizing a dataset of 5,000 SME applications with 22 financial and behavioral indicators, the system predicts the probability of financial default.

Beyond simple binary classification, the project focuses on generating a Continuous Risk Score (1-100). This allows financial institutions to categorize applicants into granular risk tiers (Low, Medium, High) based on real-world metrics like revenue volatility, NSF counts (non-sufficient funds), and debt-to-income ratios.

1.2 Objectives

- **Multi-Dimensional Risk Assessment:** Analyze 22 key features including revenue_volatility_3m, nsf_count_3m, and business_age_months to determine SME viability.
- **Predictive Scoring Engine:** Implement a machine learning pipeline that outputs a Risk Score (1-100) using probability estimation (predict_proba).
- **Explainable Financial Decisions:** Integrate SHAP to justify loan rejections, highlighting factors like "High NSF Count" or "Low Owner Credit Score" to ensure transparency.
- **Risk Dashboard:** A professional Streamlit interface featuring a Risk Score Meter and comparative SME benchmarking.

1.3 Project Scope

- **In Scope:**
  - **Preprocessing of 5,000 records including log transformations for financial variables (e.g., credit_amount_log).**
  - **Feature Selection from 22 initial variables to optimize model performance.**
  - **Training Advanced Gradient Boosting models (XGBoost/LightGBM).**
  - **Conversion of classification probabilities into a 1-100 Risk Scale.**
  - **SHAP-based interpretability module for financial transparency.**
- **Out of Scope:**
  - **Real-time integration with external credit bureaus (simulated via owner_credit_score).**
  - **Processing of raw bank statements (data is provided in aggregated CSV format).**

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2. Task Assignment & Roles

High-Performance Team Roles

Each member is assigned a specialized domain to ensure end-to-end project integrity, from raw data ingestion to explainable AI deployment**.**

Basma Bassem: Data Collection & Dataset Engineer

• **Responsibilities:** Acquire the primary dataset, perform initial validation, and generate Synthetic Data using the Faker library to handle minority class scarcity.

• **Tech Stack:** Python, Pandas, Kaggle API, Faker.

Beshoy Emad: Data Cleaning & Preprocessing Specialist

• **Responsibilities:** Implement automated pipelines to handle missing values, detect outliers in financial variables, and perform categorical encoding.

• **Tech Stack**: Scikit-learn (SimpleImputer), Numpy, Pandas, Feature-engine.

Ziad Ahmed: EDA & Business Intelligence Analyst

• **Responsibilities:** Conduct deep-dive Exploratory Data Analysis to identify hidden patterns, correlation matrices, and risk distribution visualizations.

• **Tech Stack:** Matplotlib, Seaborn, Plotly (Interactive Charts), Pandas Profiling.

Mohamed AbdElhamid: Feature Engineering & Selection Architect

• **Responsibilities:** Design and derive new financial ratios (e.g., Debt-to-Income, Liquidity Ratio) and perform dimensionality reduction to enhance model signal.

• **Tech Stack:** Scikit-learn (PCA, SelectKBest), Pandas, Category Encoders.

AbdElrahman Atef: Machine Learning & Optimization Engineer

• **Responsibilities:** Develop advanced ensemble models (XGBoost, LightGBM) and handle severe class imbalance using SMOTE or cost-sensitive learning.

• **Tech Stack:** XGBoost, LightGBM, Imbalanced-learn, Optuna (Hyperparameter Tuning).

Amr Mahmoud [Team leader] : Deployment & Explainable AI (XAI) Specialist

• **Responsibilities:** Build the Streamlit interactive dashboard, integrate model weights, and implement SHAP values to provide transparent risk explanations.

• **Tech Stack:** Streamlit, SHAP, Flask/FastAPI, Git, Docker.

3. Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Financial Outliers	High	Medium	Use robust scaling and log transformations on skewed features like credit_amount.
Model Biases	Medium	High	Audit model performance across different industries (business_industry) to ensure fairness.
False Rejection of SMEs	Low	High	Optimize thresholding to balance Recall (risk detection) and Precision.

4. KPIs (Key Performance Indicators)

KPI	Target	Measurement Method
Recall (Sensitivity)	90%>	Ensuring the model catches of high-risk SMEs (risk_clean=1).
F1-Score	0.88>	Harmonic mean for stable classification performance.
Inference Latency	150ms<	Time to process a new SME profile and return a Risk Score.
Dashboard Usability	100%	Successful rendering of the Risk Score Meter and SHAP plots.